

Power-Based Normalization of Team Performance in Mixed Indoor Rowing Relays

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We present a novel method for normalizing the results of a mixed-gender indoor-rowing relay competition based on power as performance quantifier. The Råda Rowing Challenge 2026 involved ten five-person relay teams with different gender compositions, making direct comparison of raw distances problematic. We introduce a power-based normalization scheme in which each team's average power output is compared to a composition-weighted reference derived from world indoor-rowing records. We show that this approach is physically better motivated than a linear time-based normalization, because rowing performance scales as the cube of the rowing velocity. The method is applied to the competition results, and we discuss how the ranking differs from a naive distance ranking.

I. INTRODUCTION

The Råda Rowing Challenge (RRC) is an indoor rowing competition organized by Mölnåls Roddklubb [1], held annually at RådaRum in Mölnlycke, Sweden. The 2026 edition took place on 7 March 2026 and included individual events as well as a 30-minute mixed-relay format [2]. In the relay, each team consists of five rowers who take turns on a Concept2 RowErg ergometer; the combined distance covered by all five during the 30-minute window is the team's result.

Ten teams from clubs across the Gothenburg region entered the relay: Chalmers Roddklubb (two teams), Jönköpings Roddsällskap (two teams), Halmstads Roddklubb, Mölnåls Roddklubb (two teams), Göteborgs Roddklubb, Råda Crew, and CTH-GU. Team compositions ranged from all-female (Göteborgs Roddklubb) to four-men-one-woman (Chalmers Roddklubb – Team 2), reflecting the mixed nature of the competition.

Because men and women differ systematically in their power output on the ergometer, a simple comparison of total relay distances would favour teams with more male rowers. To enable a fair ranking across different gender compositions, we develop a normalization procedure described in the following sections. Section II introduces the ergometer power formula. Section III summarizes relevant world records and the performance gap between genders. Section IV presents two normalization approaches and argues for power-based normalization. Section V reports the competition results.

II. THE ERGOMETER POWER FORMULA

II.1. Derivation

The Concept2 ergometer flywheel decelerates between drive phases due to air resistance. The drag force scales quadratically with the flywheel angular velocity, which is proportional to the

handle velocity v of the rower. Since power equals force times velocity, the average power delivered to the flywheel is [6]

$$P = cv^3, \quad (1)$$

where $v = d/T$ is the average rowing velocity (distance d in metres, time T in seconds) and c is the flywheel drag constant. For the Concept2 RowErg [3],

$$c = 2.8 \text{ W s}^3/\text{m}^3. \quad (2)$$

This value is Concept2's published calibration constant, determined empirically from flywheel drag measurements [3]. Physically, c encodes the aerodynamic properties of the flywheel and its housing; the monitor's internal algorithm normalises for damper-setting variations, rendering c effectively universal across machines. The exact machine-to-machine variation of c is not publicly documented; however, any systematic offset in c cancels identically in the normalized score $S_k = P_k^{\text{act}}/P_k^{\text{ref}}$ (Eq. (12)), provided all teams row on comparable machines. An independent validation study using force sensors found that the Concept2 monitor underestimates absolute power by approximately 25 W ($\approx 7\%$) at trained-rower intensities, with the offset being largely independent of power level [9]; because it appears in both numerator and denominator of S_k , this systematic bias cancels in the normalized score. Equation (1) can equivalently be expressed using the *500-metre split time* t_{500} , the time to row 500 m at the current average pace:

$$P = 2.8 \left(\frac{500 \text{ m}}{t_{500}} \right)^3, \quad t_{500} = \frac{500 \text{ m} \cdot T}{d}. \quad (3)$$

A faster split (smaller t_{500}) gives higher power, as physically expected. For example, the men's reference pace (2000 m in 334.7 s) corresponds to $t_{500} = 500 \times 334.7/2000 \approx 84$ s (split 1:24 min).

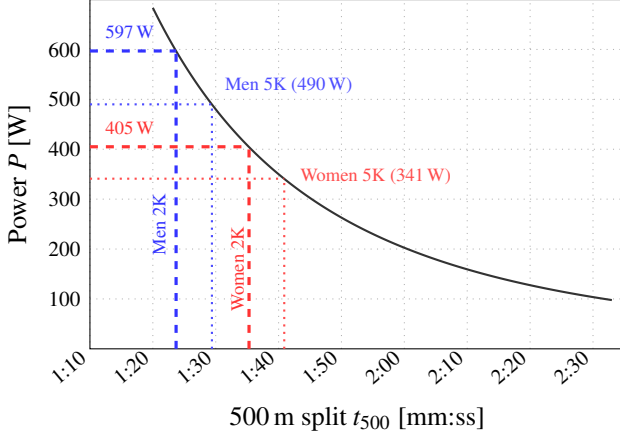


Figure 1: Ergometer power vs. 500 m split time, Eq. (3). Dashed lines: reference performance levels used in this study (2000 m world records). Dotted lines: 5000 m world records (men: T. George 2022; women: G. Rowe 2026).

II.2. Power curve

Figure 1 shows P as a function of t_{500} over the range typical for competitive indoor rowing. The strong nonlinearity – power grows as the *cube* of velocity – is the central motivation for using power rather than velocity or split time as the performance metric.

III. REFERENCE PERFORMANCE

III.1. World indoor rowing records

Table 1 lists indoor rowing records for 500 m, 1000 m, 2000 m, and 5000 m on the static Concept2 RowErg. The 500 m, 2000 m, and 5000 m entries are current world records from the Concept2 records database [4]. The 1000 m entries are event-winner times from the 2026 World Rowing Virtual Indoor Championships (WRVIC 2026) [5], which featured that distance; official world-record status for the women’s 1000 m entry has not been confirmed.

Table 1: Indoor rowing records (Concept2 RowErg, open category). Power computed via Eq. (3). 500 m, 2000 m, and 5000 m entries are current world records [4]. 1000 m entries are WRVIC 2026 event-winner times [5] (world-record status for the women’s 1000 m unconfirmed).

Dist.	Gender	Time [m:s]	P [W]	Athlete (year)
500 m	Men	1:09.8	1027	P. Clapp (2024)
500 m	Women	1:24.5	580	O. Buryak (2017)
1000 m	Men	2:38.0	710	A. Panizza (2026)
1000 m	Women	3:09.4	412	A. Ader (2026)
2000 m	Men	5:34.7	597	O. Zeidler (2026)
2000 m	Women	6:21.1	405	B. Mooney (2021)
5000 m	Men	14:53.9	490	T. George (2022)
5000 m	Women	16:49.4	341	G. Rowe (2026)

III.2. Performance spread and the gender gap

To characterise the performance spread, we define the top-athlete range as world-record power down to 80 % of that value. For men this gives 478 W–597 W (2000 m times 5:35–5:59); for women the corresponding range is 324 W–405 W (6:21–6:50). The absolute power gap between the two reference levels in Table 1 is about 47 % (597 W vs. 405 W for 2000 m). In split time, the same gap is only 14 % (1:23.7 vs. 1:35.3 per 500 m). This asymmetry is a direct consequence of the cubic relation $P \propto v^3$: a moderate advantage in speed translates into a much larger advantage in power. As a result, the gender composition of a relay team has a strong systematic effect on the expected power output, motivating the normalization approach described in the next section.

III.3. Power as performance measure

The normalization requires a choice of reference distance (and associated world record). In a 30-minute relay with five rowers, each member accumulates roughly 6 minutes of rowing, but not as a continuous effort: teams typically swap every 30 s or so, meaning each rower completes approximately 12 sprint bursts of 30 s, covering a total distance of well under 2000 m. The 2000 m world record, corresponding to an approximately 6-minute all-out effort, therefore matches both the total accumulated distance per rower and the overall duration of each rower’s contribution to the relay most closely. We therefore adopt the 2000 m world record as the normalization reference throughout (Section V), and discuss the sensitivity to this choice below.

To quantify the sensitivity to this choice, we compute the male/female reference-power ratio at all four available distances (Table 1):

$$\frac{P_m^{500}}{P_f^{500}} = \frac{1027}{580} \approx 1.77, \quad (4)$$

$$\frac{P_m^{1k}}{P_f^{1k}} = \frac{710}{412} \approx 1.72, \quad (5)$$

$$\frac{P_m^{2k}}{P_f^{2k}} = \frac{597}{405} \approx 1.47, \quad (6)$$

$$\frac{P_m^{5k}}{P_f^{5k}} = \frac{490}{341} \approx 1.44. \quad (7)$$

The male/female power gap decreases monotonically with distance, reflecting the greater male advantage in short, anaerobic efforts relative to longer aerobic exercise. Choosing a shorter-distance reference therefore sets a more demanding standard for male rowers (and conversely favours female-heavy teams), while a longer-distance reference narrows the gap. Notably, the 2000 m and 5000 m ratios (1.47 and 1.44) are very close to each other – a relative difference of only $\sim 2\%$ – whereas the step from 1000 m to 2000 m is considerably larger ($\sim 16\%$). This suggests that the normalization is relatively insensitive to the choice between 2000 m and 5000 m references, which is reassuring given the ambiguity in the appropriate reference distance for the relay format.

IV. NORMALIZATION

Let team k comprise n_m male and n_f female rowers, with $n = n_m + n_f$. During the 30-minute relay ($T = 1800$ s) the team covers a total distance D_k , giving an average team power

$$P_k^{\text{act}} = 2.8 \left(\frac{D_k}{T} \right)^3. \quad (8)$$

To normalize this total distance we consider two approaches: a time-based and a power-based normalization.

IV.1. Time-based normalization

A natural first approach is to compare D_k to the distance a reference team of the same composition would cover at the gender-specific reference pace. Defining $v_m = d_{\text{ref}}/T_m^{\text{ref}}$ and $v_f = d_{\text{ref}}/T_f^{\text{ref}}$ as the male and female reference velocities, the expected relay distance is

$$D_k^{\text{ref,t}} = \frac{T}{n} (n_m v_m + n_f v_f), \quad (9)$$

The time-normalized score

$$S_k^{(t)} = \frac{D_k}{D_k^{\text{ref,t}}} \quad (10)$$

is dimensionless and equals 1.0 for a team rowing exactly at the composition-weighted reference pace.

IV.2. Power-based normalization

We propose instead to compare power outputs directly. The reference power for team k is the composition-weighted mean of individual reference powers:

$$P_k^{\text{ref}} = \frac{n_m P_m + n_f P_f}{n}, \quad (11)$$

where P_m and P_f are the male and female reference powers (from Table 1, rows 5–6). The power-normalized score is

$$S_k = \frac{P_k^{\text{act}}}{P_k^{\text{ref}}}. \quad (12)$$

IV.3. Why power normalization is preferred

The two methods treat the nonlinearity of $P \propto v^3$ differently. Time normalization is linear in velocity: it credits a rower who rows 10 % above reference with a 10 % bonus, regardless of the absolute speed difference. Power normalization is cubic: the same 10 % velocity bonus corresponds to $(1.10)^3 - 1 \approx 33$ % more power, and this full advantage is reflected in the score.

More importantly, the Concept2 ergometer *measures* power – it is the physically native quantity of the instrument. Averaging power across team members, Eq. (11), correctly weights each rower’s physical contribution. By contrast, averaging reference velocities and then converting to a reference distance, Eq. (9), applies a linear average where the true relationship is cubic, underestimating the advantage of having high-power rowers on the team.

Power-based normalization also has a natural interpretation: a score of $S_k = 1.0$ means the team delivered exactly the power expected for their composition; $S_k > 1$ means they exceeded the reference, and $S_k < 1$ that they fell short.

Figure 2 illustrates how the reference performance depends on team composition and the choice of reference distance. Panels (a) and (b) show the composition-weighted reference time and power for all six possible compositions of a five-person team (5M/0W through 0M/5W) across the four available record distances (500 m, 1000 m, 2000 m, 5000 m). Reference times increase monotonically with distance (panel a) and are nearly linear in the fraction of female rowers at each distance. Reference powers decrease monotonically with distance (panel b), and the gap between the most male-heavy (5M/0W) and most female-heavy (0M/5W) compositions narrows as the distance grows: the male/female power ratio falls from 1.77 at 500 m to 1.44 at 5000 m, reflecting the greater male advantage in short anaerobic efforts.

Importantly, the 2000 m and 5000 m reference powers lie very close to each other (ratio 1.47 vs. 1.44, a $\approx 2\%$ difference), whereas the step from 1000 m to 2000 m is $\approx 16\%$. This insensitivity to the 2000 m vs. 5000 m choice is visible across all six compositions in panel (b), and makes the 2000 m selection robust.

Panel (c) shows both reference time and reference power at the chosen 2000 m distance as a function of team composition. Reference time varies nearly linearly from 5:34.7 (5M/0W) to 6:21.1 (0M/5W), because time averaging is linear in the fraction of female rowers. Reference power, by contrast, decreases more steeply from 597 W (5M/0W) down to a minimum of 370 W for a 1M/4W team, then rises slightly to 405 W for the all-female 0M/5W team – a direct consequence of the cubic relation between velocity and power: the all-female world record holder rows faster than the linear interpolation of mixed-team reference paces would predict, and this faster pace translates into a disproportionately higher reference power. This non-linearity is exactly what the power-based normalization captures, and what a time-based approach would miss.

V. RESULTS

V.1. Reference parameters

We use the 2000 m world records [4] (Table 1, rows 5–6) as reference performance, giving $P_m = 597$ W and $P_f = 405$ W. The 2000 m distance matches the total distance accumulated by each relay member (6 minutes at competitive club pace, see Section III.3).

V.2. Race results

Table 2 presents the ten teams, their compositions, the total relay distances, reference and actual power, and normalized scores computed via Eqs. (8)–(12). Figure 3 visualizes the same data.

V.3. Discussion

All scores are below 1.0, as expected when using world records as the reference: no relay team rows at the pace of the current world-record holder. The scores therefore represent the fraction of world-record power output delivered per rower for the given team composition.

Across all ten teams, the mean total distance was (8.60 ± 0.76) km (mean $\pm$ sample std. dev., coefficient of variation 8.8%). The mean normalized score was 0.640, meaning that teams collectively produced approximately 64% of the world-record power expected for their composition. Despite the comparatively small spread in raw distances, the score range is much wider (0.336–0.794), because the cubic power law amplifies pace differences into larger performance gaps: a team covering 550 m less than average (one standard deviation) corresponds to a pace deficit of only $\approx 3\%$, yet a power shortfall of $\approx 9\%$.

The power-normalized ranking differs substantially from a raw-distance ranking for several teams. CTH-GU covered the greatest total distance (9.511 km) and also ranks first, combining high output with a three-man composition ($P_{\text{ref}} = 520$ W). Chalmers Roddklubb – Team 1 ranks second with a two-man composition ($P_{\text{ref}} = 482$ W) and a slightly lower absolute distance. Göteborgs Roddklubb – the only all-female team – ranks third, having delivered 74.8% of world-record power against the lowest reference ($P_{\text{ref}} = 405$ W).

Chalmers Roddklubb – Team 2 illustrates the penalty for a heavy-male composition: despite covering 9.095 km (fourth largest raw distance), they rank seventh after normalization, because their four-man composition sets a demanding reference bar of 559 W.

The cubic power law amplifies pace differences into larger score differences. The rank-1 team (score 0.794) averaged approximately 1:34.6 per 500 m against a world-record reference of 1:27.7 per 500 m – a pace deficit of only $\sim 7\%$, yet a power shortfall of $\sim 21\%$.

VI. CONCLUSION AND OUTLOOK

We have presented a novel power-based normalization method for mixed-gender indoor rowing relay competitions. The fundamental argument is that rowing power scales as the cube of velocity, Eq. (1), so small velocity differences between genders translate into large power differences. Normalizing by power rather than by distance or time therefore provides a physically motivated and instrument-native measure of relative team performance.

Applied to the RRC 2026, the normalized ranking differs substantially from the raw-distance ranking for several teams. The correction is particularly significant for all-female and predominantly male teams, illustrating the importance of accounting for gender composition in mixed relay events.

The normalization algorithm is released as the open-source Python package *ergrace* [8] and can be applied to future editions of the competition or similar mixed-relay formats.

Our proposed method can be further extended and applied to different age and weight groups, including official handicap categories as defined by World Rowing [7]. Incorporating age-graded reference powers and lightweight classifications would allow fair comparison across an even broader range of team compositions, opening the approach to wider use in community and masters rowing events.

DATA AVAILABILITY

The race data, normalization code, and the source of this paper are openly available in the *ergrace* package repository [8].

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AUTHOR CONTRIBUTIONS

LJS wrote the paper and developed the power normalization. JL assisted with data acquisition and analysis. OG acquired the data and supervised the project.

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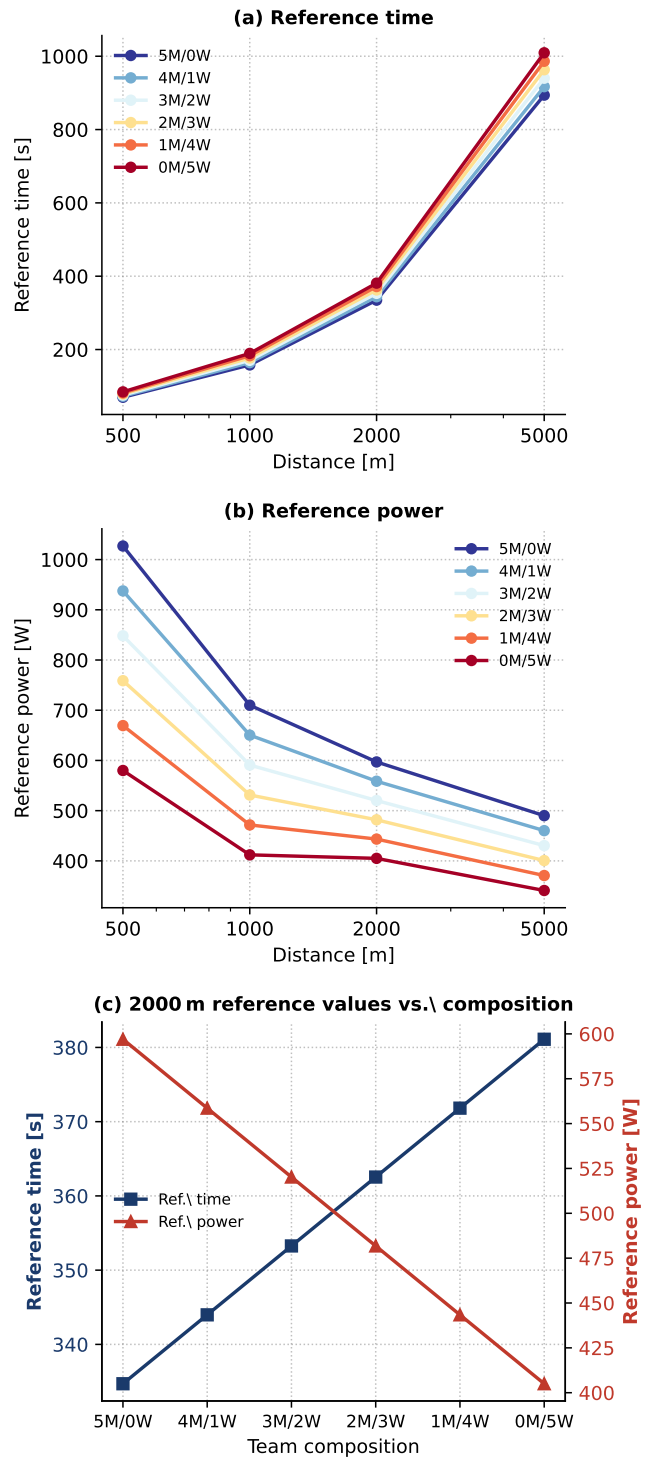


Figure 2: Reference performance as a function of event distance and team composition for all six possible five-person gender mixes. (a) Composition-weighted reference time T_k^{ref} vs. distance (log scale), showing the monotonic increase with distance and with the fraction of female rowers. (b) Corresponding reference power P_k^{ref} vs. distance; note the convergence of all compositions towards shorter distances and the near-identical values at 2000 m and 5000 m. (c) Reference time (left axis, navy) and reference power (right axis, red) at the chosen reference distance of 2000 m, plotted against team composition. The non-monotonic dip in power at 1M/4W – below the all-female value – reflects the cubic velocity-to-power conversion and motivates the power-based normalisation.

Table 2: RRC 2026 relay results ranked by normalized score. n_m/n_f : male/female rowers. P_{ref} : composition-weighted reference power, Eq. (11). P_{act} : team average power, Eq. (8). Score = $P_{\text{act}}/P_{\text{ref}}$.

Rank	Team	Lane	Comp.	Dist. [km]	P_{ref} [W]	P_{act} [W]	Score
1	CTH-GU	10	3M/2F	9.511	520	413	0.794
2	Chalmers Roddklubb – Team 1	8	2M/3F	9.268	482	382	0.793
3	Göteborgs Roddklubb	7	0M/5F	8.575	405	303	0.748
4	Jönköpings Rodds. – Team 2	5	2M/3F	9.021	482	353	0.733
5	Mölnåls Roddklubb – Team 1	4	2M/3F	8.744	482	321	0.666
6	Mölnåls Roddklubb – Team 2	6	1M/4F	8.439	443	289	0.652
7	Chalmers Roddklubb – Team 2	1	4M/1F	9.095	559	361	0.646
8	Jönköpings Rodds. – Team 1	2	3M/2F	8.660	520	312	0.600
9	Råda Crew	9	3M/2F	7.733	520	222	0.427
10	Halmstads Roddklubb	3	2M/3F	6.956	482	162	0.336

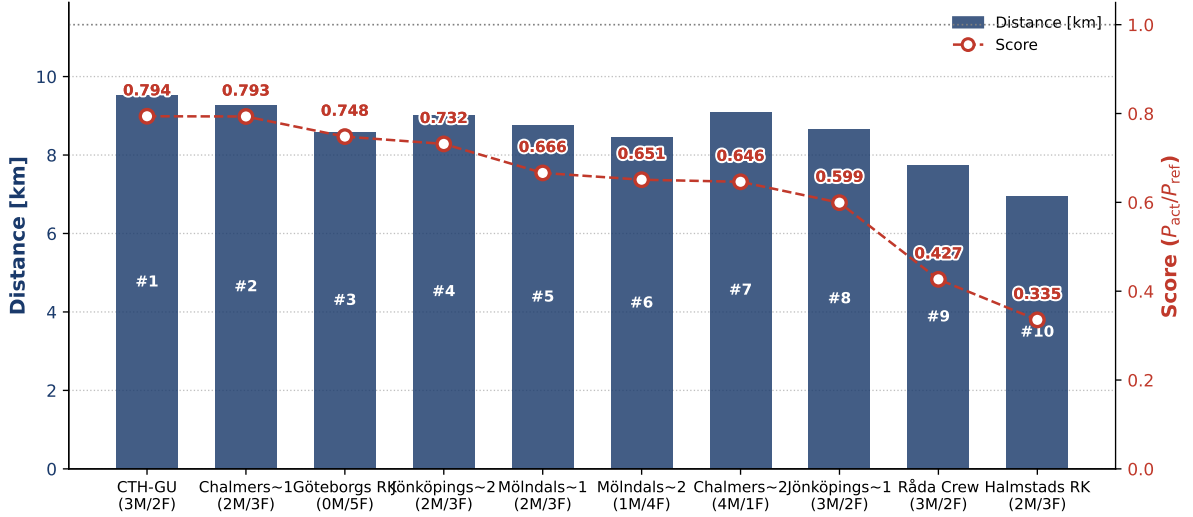


Figure 3: RRC 2026 relay results. Bars (left axis): total relay distance per team. Markers (right axis): normalized score (fraction of 2000 m world-record power for the team composition). Teams ordered by score (best to worst, left to right). Generate this figure by running `generate_figures.py`.